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The effects of rainfall and evapotranspiration on the temporal dynamics of *Anopheles gambiae* s.s. and *Anopheles arabiensis* in a Kenyan village

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Abstract

The population dynamics of the larval and adult life stages of the malaria vector *Anopheles gambiae* Giles were studied in Miwani, western Kenya, in relation to meteorological conditions. Larval density within a habitat, the number of larval habitats and sibling species composition were investigated as determinants of larval population dynamics. Female vector densities inside local houses and sibling species composition were investigated as determinants of adult population dynamics.

Larval densities were estimated using a modified area-sampling method. Within the habitats, all instars showed a highly aggregated distribution, with the exception of second instars. A longitudinal study on the larval populations of *A. gambiae* s.l. in two different types of habitat (dirt track and ditch) was carried out, using a novel sampling procedure. *A. gambiae* s.s. and *Anopheles arabiensis*, the two sibling species occurring sympatrically in the study area, showed some spatial segregation between the two types of habitat. Rainfall was significantly correlated with the number of *A. gambiae* s.l. larval habitats during the first 6 weeks of study taking 1 week time lag into account, while over the entire 5-month study period correlations were less clear. With 1 week time lag, rainfall was also significantly correlated with the number of female *A. gambiae* s.l. collected from CDC-light traps in the study houses. Both larval and adult populations showed a significant increase in the proportion of *A. gambiae* s.s. within the mixed population of *A. gambiae* s.s. and *A. arabiensis* over time. Although not significantly correlated, the ratio of rainfall over precipitation/potential evapotranspiration (P/PE), indicative of the humidity conditions in the area, was probably the driving force of this increase.
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Keywords: *Anopheles gambiae* s.s.; *Anopheles arabiensis*; Malaria mosquito; Larval density; Population dynamics; Rainfall; Temperature; Potential evapotranspiration; Kenya

1. Introduction

Since the discovery of *Anopheles gambiae* as a complex of several cryptic sibling species (Davidson, 1962), many studies have been undertaken to understand the distribution, behaviour and population

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dynamics of this vector of malaria in sub-Saharan Africa. Population dynamic studies have been conducted for a variety of reasons, such as gaining a basic understanding of malaria transmission parameters (Molineaux and Gramiccia, 1980), assessing the impact of insecticide control measures (Service et al., 1978) and identifying suitable areas for future vaccine trials (Petrarca et al., 1991).

A. gambiae Giles s.s. and *Anopheles arabiensis* Patton, the two most common members of the *A. gambiae* complex, share habitats in places where their distributions overlap. Both species can be found in small, transient habitats and have similar requirements for their larval environment (Service, 1970; Service et al., 1978; Gimnig et al., 2001). Minakawa et al. (1999) found a significant spatial heterogeneity within the Suba District of western Kenya, although such a heterogeneity was not apparent in other studies of a similar sized area (White and Rosen, 1973; Service et al., 1978). Temporal heterogeneity in species composition was detected in Mali when different years were compared (Edillo et al., 2002). Although *A. arabiensis* has been found in association with pools close to cattle (Charlwood and Edoh, 1996) and the abundance of *A. gambiae* s.s. has been associated with distance to the nearest house (Minakawa et al., 1999, 2002), no key environmental variables causing spatial heterogeneity in the larval population have been identified.

Studies that have investigated changes in larval species composition on a more continuous time scale have been able to relate changes in the abundance of the two species to rainfall patterns. Gimnig et al. (2001) found that the proportion of larvae of *A. gambiae* s.s. relative to that of *A. arabiensis* was much greater during and immediately after the rains compared with drier periods in western Kenya.

Similar to the larval population, the relative proportion of *A. gambiae* s.s. in the adult population peaks shortly after the onset of the rainy season, while *A. arabiensis* is the more dominant species during the dry season (White et al., 1972; Petrarca et al., 1987; Githeko et al., 1996a; Lemasson et al., 1997; Awolola et al., 2002). A comparable differentiation can be observed on a spatial scale with *A. arabiensis* occurring in the more arid areas of Africa (Gillies and Coetzee, 1987; Petrarca et al., 2000). On a finer spatial scale, adult *A. gambiae* s.s. were collected more frequently in the foothills of western Kenya, while *A. arabiensis* was more abundant on the plains that received less rain (White, 1972; Highton et al., 1979). This difference has been explained by the different sensitivities of *A. gambiae* s.s. and *A. arabiensis* towards humidity conditions (Coz, 1973). Lindsay et al. (1998) were the first to describe mathematically the relationship between humidity conditions and relative abundance of the two species, using field data from West Africa.

Information gained from all these and other studies has been used to compile distribution maps of the two sibling species over the African continent (Coetzee et al., 2000; Rogers et al., 2002). Using climate data to model the ranges of the two sibling species on the African continent, Lindsay et al. (1998) found good agreement between predicted and actual relative abundance, indicating that climate (temperature and rainfall) is the driving force of the spatial distribution of *A. gambiae* s.l.

Although our knowledge on the population dynamics of *A. gambiae* s.l. has been upgraded to the continental scale, no attempts have been made to perform a similar upgrading on a temporal scale. In the near future, it is expected that climatic and environmental changes, such as global warming, land use change and construction of water control systems, will affect malaria risk considerably (Patz et al., 2000; Githeko et al., 2000). Therefore, it will be important to identify forces that drive the proliferation of the two species as a result of these changes. Substantial changes over time in the ratio of the two species will affect malaria risk because *A. gambiae* and *A. arabiensis* differ in their ability to transmit malaria as a result of their different blood-feeding preferences (Githeko et al., 1994).

Currently, an integrated study on environmental change and malaria transmission risk, covering both spatial and temporal scales, is being performed in western Kenya. As part of that study, we investigated the population dynamics of *A. gambiae* s.s. and *A. arabiensis* in the larval and adult stages. During the field study, we focussed on the density of larvae within a habitat, the number of larval habitats and the sibling species ratio within the habitats, since these are the main parameters expressing larval population dynamics. The larval data were compared with the adult anopheline density inside study houses and with the species ratio of this population in order to assess

the relationship between the aquatic and terrestrial populations.

2. Materials and methods

2.1. Study area

The studies were carried out in Miwani, a small rural village situated on the Kano Plain in western Kenya, 30 km east of Kisumu (0°06′S, 34°45′E; Fig. 1). The Kano Plain is elevated approximately 1200 m above sea level and is covered with fertile alluvial deposits carried there by rivers from the surrounding highlands after a heavy rain. Mean annual rainfall is 1441 mm, with a main peak of rainfall between March and May and a smaller peak of rainfall in November and December. Malaria in the study area is holo-endemic with a prevalence of *Plasmodium falciparum* of 70–90% in children (Githeko et al., 1996b).

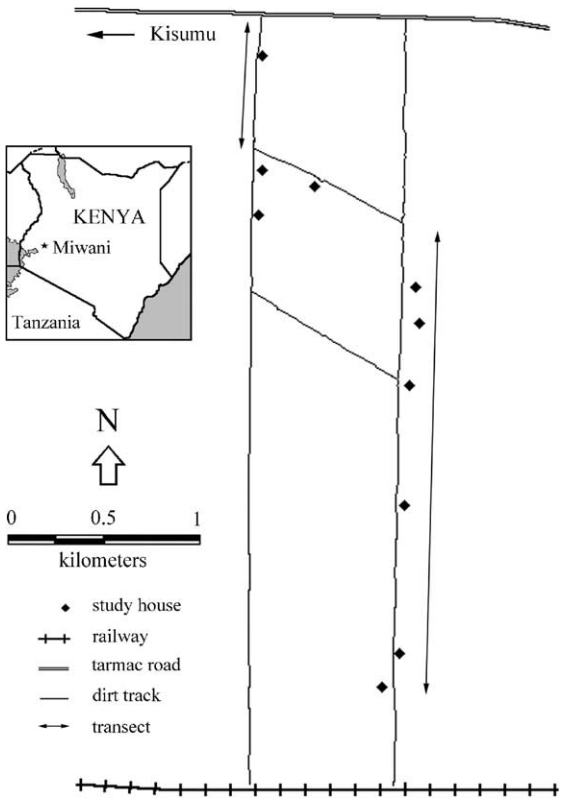


Fig. 1. Map of the study area in western Kenya.

Precipitation, temperature and radiation data for the area were obtained from the Kenya Sugar Research Foundation (KESREF) in Kibos, situated 25 km west of the study site. Radiation and temperature data were used to calculate potential evapotranspiration (PE), using the Priestley–Taylor method (Shuttleworth, 1993). This method has proven to be a good approximation of evapotranspiration in modelling soil moisture in relation to malaria transmission factors (Patz et al., 1998). The precipitation/potential evapotranspiration index (P/PE) was used to evaluate the moisture conditions in the area. A P/PE index <1 indicates drying conditions, since in that case potential evapotranspiration is higher than precipitation. Similarly, a P/PE index >1 indicates that the area gets more wet.

2.2. Larval densities

To determine the density of larvae within a habitat, we selected 18 sites containing *A. gambiae* s.l. larvae on one of the dirt tracks in the study area. The selected sites were typical for the area, since the vast majority of *A. gambiae* breeding sites can be found on these dirt tracks. All sites were shallow and transient with little to no vegetation and most of them were created through human activity (digging, driving vehicles). The size of the sites ranged from 0.31 to 3.1 m^2. Along the length axis of each site, plastic cylinders (25 cm height; 10.2 cm ∅) were put firmly on the bottom of the site at distances of 0.5 m (Fig. 2). These plastic cylinders were open on both sides and had three wooden sticks attached to one of the open ends to fix the cylinder in the soil. The number of larvae trapped in the cylinders was then counted and scored per instar. The size of the site was measured and the density of larvae estimated. The absolute number of larvae present in the site was determined by removing all larvae by hand with a plastic pipette. In total, estimated and absolute densities were compared for nine *A. gambiae* s.l. sites that were sampled between 24 June 1999 and 27 August 1999 and nine *A. gambiae* s.l. sites that were sampled between 17 April 2000 and 23 August 2000.

2.3. Habitat dynamics

In Miwani, a transect of 1.66 km dirt track and 1.66 km ditch, running along the west side of the dirt

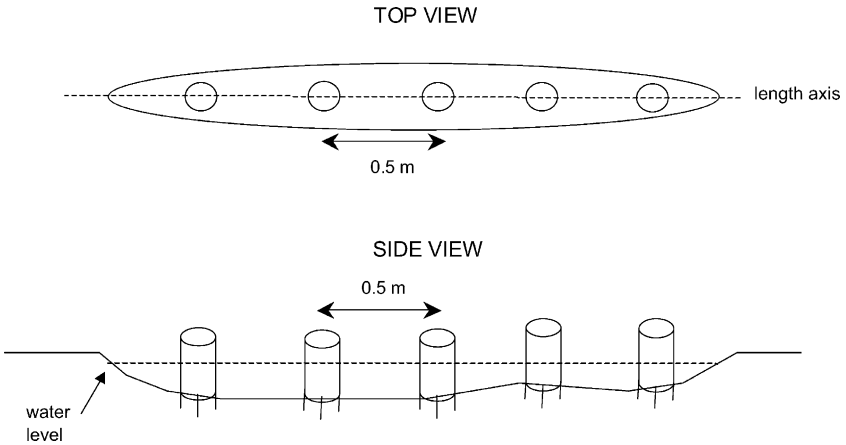


Fig. 2. Schematic representation of the positioning of the sampling cylinders in a habitat.

track, was selected (Fig. 1). For logistical reasons, this transect was sampled weekly during the first 6 weeks and every 2 weeks during the last 14 weeks, between 13 March 2000 and 22 August 2000. The first sampling of the transect coincided with the first rains of the season, although rainfall during all study months was below average and the total rainfall anomaly from March 2000 to August 2000 was −280 mm. Sampling was always conducted between 8:00 and 12:00 a.m.

All aquatic habitats on the transect were inspected for the presence of *Anopheles* larvae. An aquatic habitat was considered a 'positive' site if at least one *Anopheles* sp. larva could be found. The following methods were used to sample the larvae:

1. *Vision*. Small, shallow (generally less than 10 cm depth) habitats without or with little vegetation were visually inspected for a maximum of 5 min by two persons.
2. *Dipping*. Habitats with much vegetation were dipped for the presence of larvae (only if the habitat was deep enough to use the dipper).
3. *Using a net*. Large, deep (generally more than 10 cm depth) habitats were sampled, using a net with a plastic collection bottle (Service, 1993). The entire surface area was sampled one or two times.

If a site was positive, the larva first encountered was picked and placed in a 1.5 ml Eppendorf tube filled with approximately 1 ml of water, originating from the site. By picking one single larva from the 'positive' site and by doing this for all sites at a given moment in time, a sample is obtained, that reflects the species ratio of the adult population (i.e. their mothers).

Of each site, the size, depth and surface temperature (in the sun and/or in the shade) was recorded. Aquatic habitats were marked as 'dirt track' or 'ditch' habitats. Size was estimated in six classes: <0.5, $0.5-1$, 1–2.5, 2.5–5, 5–10 and 10–25 m^2.

On arrival in the laboratory, third and fourth instar larvae were identified to species under a microscope, using the key of Gillies and Coetzee (1987). First and second instar larvae were reared individually in cups to third or fourth instar larvae before identification. Larvae of the *A. gambiae* complex were stored individually in Eppendorf tubes with silicagel. DNA was extracted from the larvae using the extraction method described in Schneider et al. (2000). Subsequently, sibling species were determined using the species-diagnostic rDNA-PCR developed by Scott et al. (1993).

Parallel to the larval study, a longitudinal study on indoor adult vector densities was carried out in the same village, which began in September 1999. For this purpose, ten study houses were selected along the same transect (Fig. 1). CDC-light traps (Model 1012, John W. Hock Company, Gainesville, FL) were installed at the foot end of the beds of these houses (Mboera et al., 1998), near the top of a bednet, every 2 weeks and these traps run from 6:00 p.m. to 6:00 a.m. in each house. Collected mosquitoes were placed individually in Eppendorf tubes with silica gel and members of the *A. gambiae* complex were further

identified by rDNA-PCR (Scott et al., 1993). Trap data were used for the studies of adult vector densities and species composition.

2.4. Statistics

We tested if distributions of larvae within habitats fitted either a Poisson distribution, indicating a random distribution of larvae, or a negative binomial distribution, indicating an aggregated distribution. The maximum likelihood method was used to calculate the *k*-values of the negative binomial distribution, whereby smaller values of *k* indicate a higher degree of aggregation (Elkinton, 1993). Linear regression was used to test the accuracy of the estimated densities in the larval habitats.

Spearman's rank correlation coefficients were calculated between rainfall, the number of *A. gambiae* s.l. habitats and the average density of mosquitoes in the houses with no time lag, 1 week lag and 2 weeks lag, respectively. For the analysis of the proportions of *A. gambiae* s.s., collected during the sampling, we used logistic regression. Differences in continuous variables between the two types of aquatic habitat (dirt track and ditch) were compared using Mann–Whitney *U* and Kruskall–Wallis tests. Frequencies were tested with χ^2-tests.

All statistical tests were performed within the SPSS 11.0 software package.

3. Results

3.1. Larval density

Averages and variances of densities within the sample cylinders were 0.40 ± 1.38, 0.21 ± 0.56, 0.49 ± 1.35 and 0.15 ± 0.47 larvae per sample cylinder for first, second, third and fourth instars, respectively. These high variance to mean ratios indicated that the distributions of the four different instars were non-uniform. Expected densities for the Poisson distribution were significantly different from the observed densities (χ^2-analysis, $P < 0.05$), indicating that a Poisson distribution did not describe the observed pattern accurately. However, distributions of first, third and fourth instar larvae were accurately described by the negative binomial distribution, while the distribution of second instar larvae did not fit the negative binomial (χ^2-analysis, $P < 0.05$). The estimated *k*-values were 0.11, 0.48, 0.33 and 0.35 for first, second, third and fourth instars, respectively, indicating that the highest aggregation occurred in first instars, followed by third, fourth and second instars.

Estimated densities within the 18 selected larval habitats ranged from 0 to 625 larvae/m^2, with a median of 98 larvae/m^2, while absolute densities ranged from 13 to 722 larvae/m^2, with a median of 90 larvae/m^2. The proportion of the water surface that was sampled to obtain the estimates ranged from 2.5 to 16.3%, with an average of 4.8%. There was no evidence that sampling a larger surface of the larval habitat improved the estimate of density ($r_s = 0.413$, $P > 0.05$). The predictive value of the sampling method was highly significant (linear regression, $P < 0.001$, $R^2 = 0.961$, Fig. 3).

3.2. Number of larval habitats

During the study period, a total of 1045 aquatic habitats were encountered of which 604 (58%) were found positive for *Anopheles* larvae (Table 1). With increasing size of the habitat, the proportion positive for *Anopheles* larvae rose from 45% for habitats smaller than 0.5 m^2 to 78% for sites between 5 and 10 m^2. Of the largest habitats (10–25 m^2), 56% were occupied throughout the study. Chi-square analysis indicated that the proportion of sites occupied by *Anopheles* larvae differed significantly over time ($P < 0.001$) and that ditch sites were more likely to be occupied by *Anopheles* larvae than dirt track sites ($P < 0.01$). Ninety-three percent of the collected larvae could be identified to species, while the rest were unidentifiable due to death before reaching the third or fourth instar. Of the identified larvae, 94.5% were *A. gambiae* s.l., while the rest were identified as *A. coustani* Laveran (3.9%), *A. pharoensis* Theobald (1.1%) and *A. funestus* Giles (0.5%). Of all collected *A. gambiae* s.l. larvae, 72.6% were collected from dirt track sites and the remaining 27.4% were collected from the ditch. Ditch sites were deeper and covered with more vegetation than dirt track sites. Water temperatures ranged from 14.6 to 35.1 °C and from 17.1 to 36.5 °C for ditch and dirt track sites, respectively. Significant differences in temperature between ditch and dirt track sites were found for sites sampled

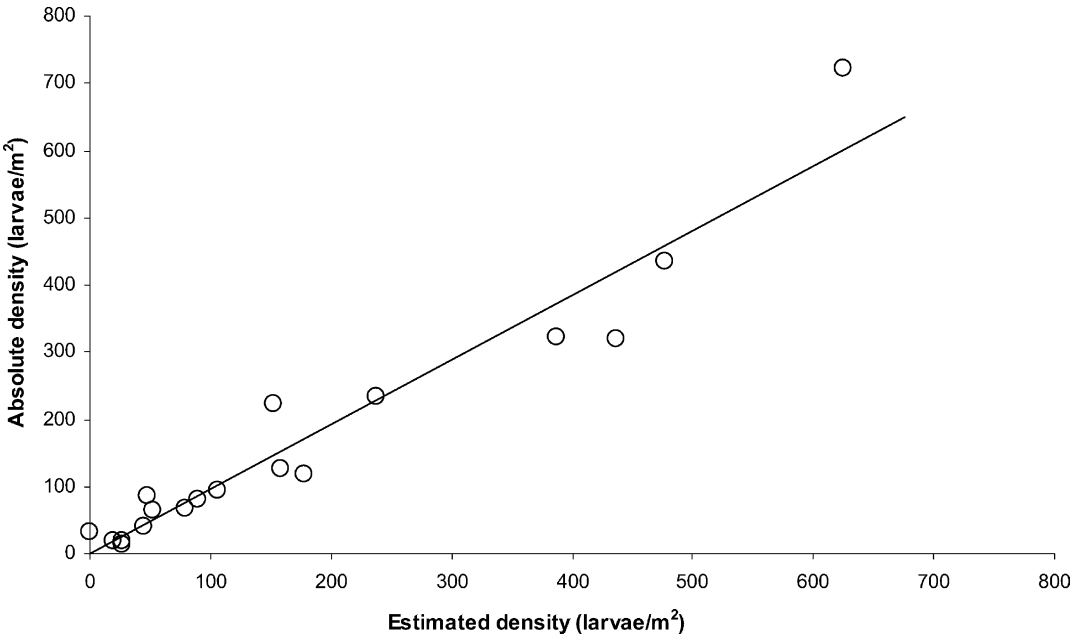


Fig. 3. Estimated density versus absolute density for the 18 selected *A. gambiae* s.l. breeding sites on the dirt tracks in Miwani.

Table 1
Summary of the number of larval habitats encountered on the transect, species identified from these habitats and characteristics of these habitats

	Dirt track	Ditch
Total encountered	743	302
Total encountered with *Anopheles* spp. present	408 (55)	196 (65)
Anopheles species identified		
A. pharoensis	3	3
A. coustani	1	21
A. funestus	0	3
A. gambiae s.l.	376	154
Sibling species identified (PCR)		
A. gambiae s.s.	75 (22)	16 (11)
A. arabiensis	263 (78)	124 (89)
Temperature (°C)		
Mean—shade (±S.D.)	21.0 ± 1.9	21.5 ± 3.1
Mean—sun (±S.D.)	25.8 ± 4.2	27.9 ± 5.3
Min—shade	17.8	14.6
Max—sun	36.5	37.4
Depth (cm)	6.2 ± 5.3	10.6 ± 7.2
Temperature difference within sites (°C)		
Mean (±S.D.)	2.9 ± 2.6	3.1 ± 2.4
Min	0.0	0.3
Max	9.0	8.9

Values in parentheses are percentages.

between 9:00 and 10:00 a.m. (Mann–Whitney *U*-test, $P < 0.05$), while during the other sampling hours temperatures in both types of habitat were the same (Mann–Whitney *U*-test, $P > 0.05$). As a result of parts of the site being exposed to the sun and parts of the side being shaded, temperature differences within a site could reach 9.0 °C in dirt track sites and 8.9 °C in ditch sites with mean values of 2.9 and 3.1 °C, respectively.

Fig. 4A shows the dynamic patterns of rainfall and the number of negative sites and positive dirt track and ditch sites throughout the study period. There were no significant correlations between rainfall and the number of *A. gambiae* s.l. larval habitats when no time lag was considered (Table 2). However, rainfall was positively correlated with the total number of *A. gambiae* s.l. habitats on the dirt track and in the ditch the following week. Correlations were only significant when the first 6 weeks were analysed separately. When a time lag of 2 weeks was considered, no significant correlations were detected.

During the study period, there was a clear peak in the number of adult female *A. gambiae* s.l. and *A. funestus* collected from the light traps in the study houses (Fig. 4B). The peak of *A. gambiae* s.l. was

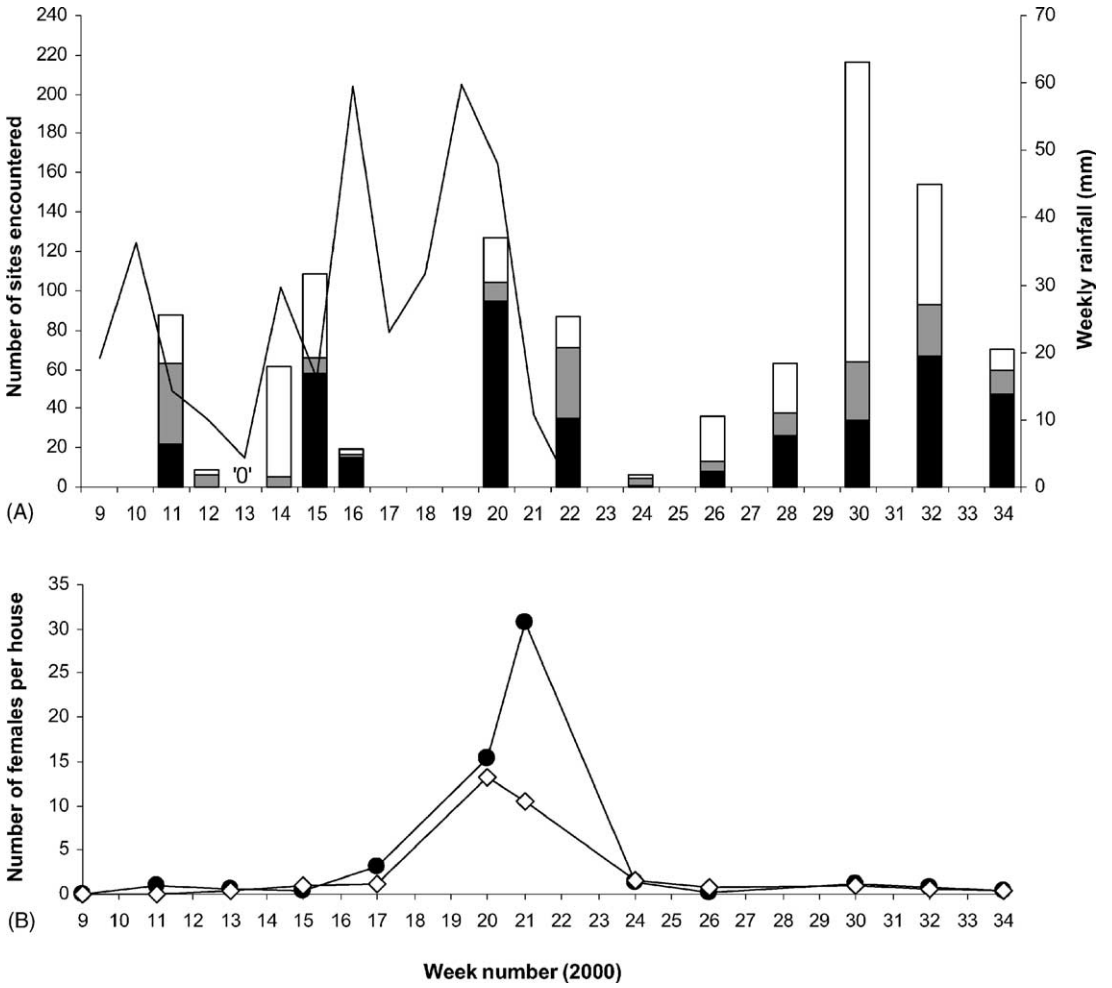


Fig. 4. (A) Weekly rainfall (solid line) and presence of anophelines in the available aquatic habitats (bars). Open bars: number of sites without *Anopheles* spp.; grey bars: the number of ditch sites with *Anopheles* spp.; black bars: number of dirt track sites with *Anopheles* spp. (B) The mean number of female *Anopheles gambiae* s.l. (solid circles) and *A. funestus* (open diamonds) collected from the light traps in the 10 selected study houses throughout the study period.

significantly correlated with rainfall, when a time lag of 1 week was taken into account (Table 2). This correlation was not significant ($P = 0.061$) when a time lag of 2 weeks was taken into account. No significant correlations were found between rainfall and the numbers of *A. funestus* ($P > 0.05$).

Because of the irregular sampling interval in the larval population, no correlations could be calculated between the number of *A. gambiae* s.l. larval habitats and the number of collected adults of this species with 1 week lag. However, if we interpolated the number of females for the weeks in which no collections were done, we obtained a significant correlation ($r_s = 0.679$, $P < 0.01$) between the number of *A. gambiae* s.l. sites and the (interpolated) number of females with 1 week lag. With no time lag and with 2 weeks lag, there were no significant correlations ($P > 0.05$). Since almost no larvae of *A. funestus* were collected from the dirt track or ditch sites, no meaningful correlations between larval and adult population densities could be determined for this species.

Table 2
Spearman's rank correlation coefficients between rainfall and (1) the total number of habitats containing *Anopheles gambiae* s.l. on the dirt track, (2) the total number of habitats containing *A. gambiae* s.l. in the ditch and (3) the average number of female *Anopheles gambiae* s.l. collected in the light traps in the study houses

	Larval habitats		Adult density
	(1) Dirt track	(2) Ditch sites	(3) Light trap collections
No time lag			
First 6 weeks	0.158	−0.108	0.280
Whole period	0.132	−0.021	
>1 week lag			
First 6 weeks	0.867[a]	0.793[a]	0.585[a]
Whole period	0.497	0.195	
>2 weeks lag			
First 6 weeks	−0.020	0.270	
Whole period	0.342	0.391	0.554

[a] $P < 0.05$, two-tailed.

Of the 478 *A. gambiae* s.l. larvae identified by PCR, 387 (81.0%) were *A. arabiensis* and 91 (19.0%) were *A. gambiae* s.s. Chi-square analysis revealed that *A. gambiae* s.s. was more likely to be collected from dirt track sites than from ditch sites ($P < 0.01$) (Table 1). Logistic regression analysis indicated that there was a significant increase in the proportion of *A. gambiae* s.s. in the larval population during the sampling period ($P < 0.05$; Fig. 5A). A similar significant model was constructed for the proportion of *A. gambiae* s.s. in the adult population, using the PCR results of the light trap catches. When the models were compared, the proportion of *A. gambiae* s.s. in the larval population increased by a factor of 4.31 and in the adult population by a factor 4.42 between week 11 and 34 of the study period. The larval model was not significantly different from the adult model ($P > 0.05$).

No significant correlations between P/PE and the proportion of *A. gambiae* s.s. in the larval population were detected at any time lag up to 2 weeks. However, the moving average line (with a period of 3) shows a moistening trend (i.e. increasing P/PE index, Fig. 5B), favourable for *A. gambiae* s.s., between week 13 and 20, while after week 20 the P/PE index decreases rapidly and stabilises around 0.4.

4. Discussion

Although larvae were highly aggregated within their habitats, it was possible to accurately estimate the total number of larvae present using the area-sampling method (Service, 1993). It remains difficult, however, to determine the productivity of the sampled sites, because we do not know the survival rate of the larvae in these sites. Previous studies have shown that larval survival differs among habitats as a result of the presence or absence of predators, parasites and pathogens (Service, 1971, 1973, 1977; Aniedu et al., 1993). These studies report a survival of 2–10% and, therefore, a relatively small change in survival rate may have drastic effects on adult mosquito abundance, which stresses the unpredictable character of mosquito numbers. Recently, we reported the occurrence of cannibalism and predation by sibling species within the *A. gambiae* complex, and this too may affect larval survival and hence productivity of a site (Koenraadt and Takken, 2003).

Since most larval habitats were long and narrow, we could not test if the observed spatially heterogeneous distribution was related to aggregation along the edges of the site. It is not likely that the observed aggregation occurred around vegetation, since the majority of larval habitats was free of vegetation. These data, therefore, suggest that there is either some degree of inherent aggregation or there is aggregation around patches of food. The latter is unlikely, since *Anopheles* larvae feed on algae and other micro-organisms, occurring throughout the surface layer of a larval habitat. It should be noted here, that density and aggregation have different effects on larval populations. Aggregation may occur at low densities and at high densities larvae may not be aggregated at all. High densities may lead to a longer larval development time and smaller adults, due to competition for food sources (Lyimo et al., 1992; Gimnig et al., 2002), while aggregation may lead to more frequent interactions between larvae, resulting in cannibalism and predation (Koenraadt and Takken, 2003).

The transect sampling results showed that populations of *A. gambiae* s.l. larvae were highly dynamic. These dynamics can be attributed mainly to rainfall. The highest correlations between rainfall and the number of larval *A. gambiae* s.l. habitats were obtained when a time lag of 1 week was taken into account.

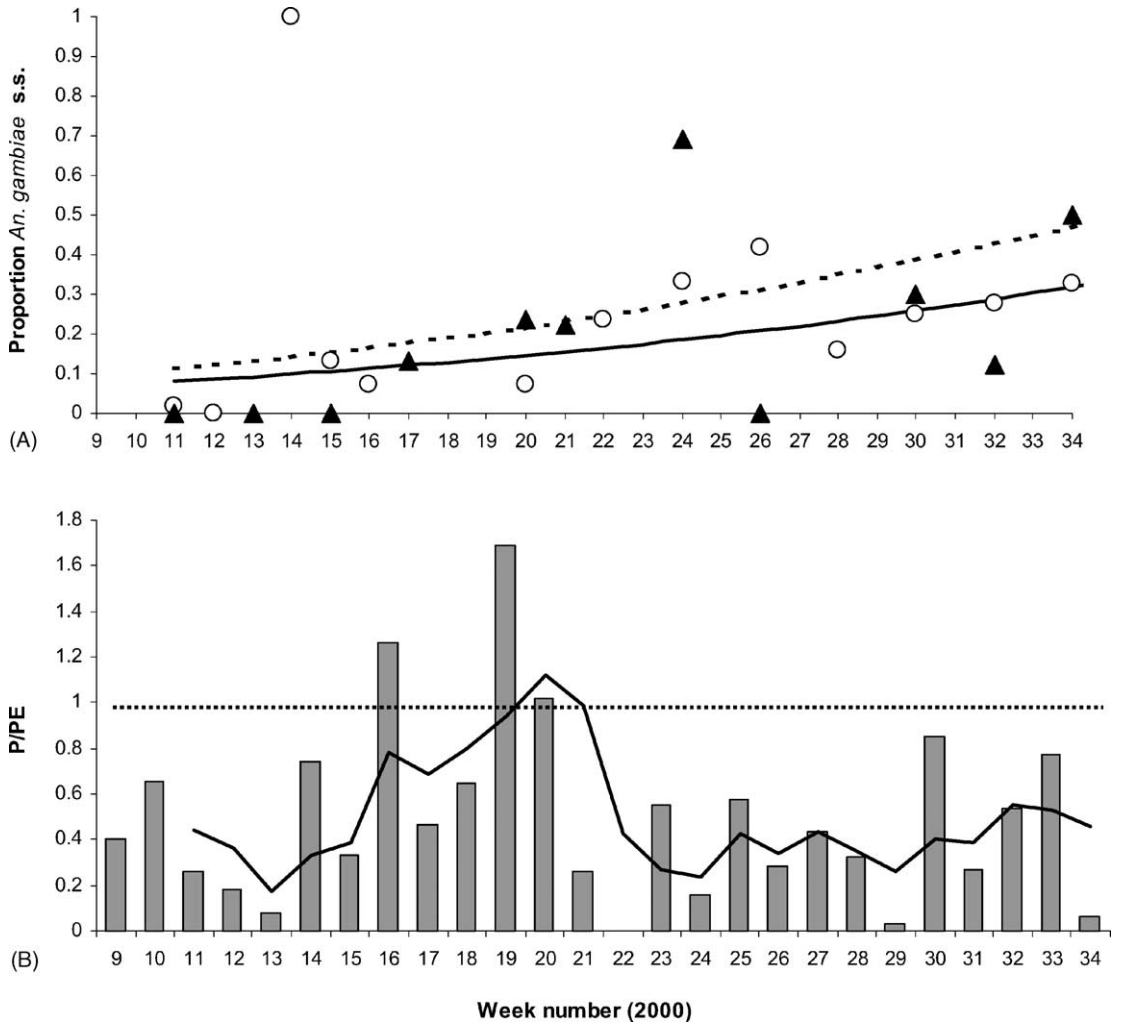


Fig. 5. (A) The fraction of *Anopheles gambiae* s.s. in the larval population (open circles) with the logistic model (solid line) and the fraction of *Anopheles gambiae* s.s. in the adult population (closed triangles) with the logistic model (dotted line). (B) The index of rainfall (P) over potential evapotranspiration (PE) as a measure of saturation deficit throughout the study period with the moving average trendline with period 3. Values above 1.0 (dotted line) indicate moist conditions.

Most probably, this reflects the time lag that exists between the formation of potential habitats by rain and the actual colonisation of these habitats by mosquitoes. With 1 week lag, there was also a significant correlation between rainfall and the number of female *A. gambiae* s.l. collected in the study houses. Since this correlation was close to significance with a time lag of 2 weeks and since there was a significant correlation between the number of *A. gambiae* s.l. habitats with the number of interpolated female *A. gambiae* s.l. with 1 week lag, these correlations most likely reflect the time it takes to colonise habitats plus the time for eggs to develop into adults. The fact that there was a drop in the number of adult *A. gambiae* s.l. while the numbers of *A. gambiae* s.l. habitats remained relatively high during the last 10 weeks of study (e.g. weeks 30 and 32), may be explained by the fact that these habitats became more permanent and were, therefore, gradually colonised by predators, parasites and pathogens, making them less productive.

The dirt track sites were more exposed to the sun and, in general, less deep than ditch sites and their number was therefore more affected by rainfall than the number of ditch sites, resulting in a higher correlation with rainfall. In addition to an effect of the type of habitat on the correlations, there was also an effect of time, since correlations were only significant when the first 6 weeks were analysed separately. Probably, once the soil was saturated after the first heavy rains (in weeks 16, 19 and 20 in this study), aquatic habitats had become more permanent and their presence will thus have been less affected by rainfall. Besides saturation of the soil as a direct effect of rainfall, another important factor determining the relation between rainfall and larval habitats is the pattern of rainfall: one heavy rain shower during a week will have a different effect than several smaller showers during the same week. Finally, the fact that rainfall data were collected 25 km from the study site may have affected correlations, since rainfall patterns may differ between the study area and the area from which the rainfall data originate. Nevertheless, weekly rainfall will probably show similar trends for both areas, because the weather station is located at the same altitude and at the same distance (∼2 km) from an escarpment. Therefore, as mentioned by Craig et al. (1999), a direct and predictable relation between rainfall and vector abundance may be hard to obtain.

Although water temperature did not differ between the two types of habitat, we observed major differences in water temperature, up to 9 °C, within a habitat. Optimal temperature for development of *A. gambiae* s.s. lies around 27 °C (Lyimo et al., 1992) and to prevent potential negative effects of sub-optimal temperatures, larvae in natural habitats should be capable of migrating to that part of a site most optimal for their development. To what extent larvae express this type of behavioural adaptation to changing environmental conditions remains to be studied.

Pools and puddles encountered on the dirt track and in the ditch make up the vast majority of available habitats in the area of Miwani. Larvae of *A. gambiae* s.s. were more likely to be collected from dirt track habitats than from ditch habitats, indicative of some spatial segregation. Since both types of habitat were running parallel to each other, this preference is most likely the result of an innate preference for physical or chemical characteristics of these habitats and not a result of other environmental factors such as distance of the habitat to cattle (Charlwood and Edoh, 1996) or distance of the habitat to the nearest house (Minakawa et al., 1999, 2002).

Adult *A. funestus* were evidently present in Miwani, but rarely collected from the aquatic habitats on the transect. Given the intensity of our investigations, we suggest that *A. funestus* immigrated into the study area. Most likely, the adults originate from the large, permanent habitats in the vicinity of the Ahero rice irrigation scheme a few kilometres to the south of Miwani. Adult densities of *A. funestus* were found to be approximately 10 times higher in Ahero than in Miwani (Githeko et al., 1996a), supporting the idea of migration from areas where intense breeding occurs to areas with little to no breeding.

Temporal variation in sibling species composition may be the result of different capabilities to adapt to climatological conditions. From laboratory studies it is known that adult *A. arabiensis* perform better under conditions of low relative humidity (Coz, 1973), while larvae of *A. gambiae* s.s. are stronger competitors than *A. arabiensis* when kept in mixed populations (Schneider et al., 2000). These differences may lead to a dominance of *A. arabiensis* during the dry season, but this may shift gradually to a dominance of *A. gambiae* s.s. as soon as the rains start. Our results showed that the proportion of *A. gambiae* s.s. in the larval as well as in the adult population increased with a similar rate during the study period. This increase coincided with an increase in humidity conditions, expressed as an increase in the P/PE ratio. Our results also support the model findings of Lindsay et al. (1998), since they found that 5-month optimum P/PE values lower than 1.0 are indicative of a dominance of *A. arabiensis* and P/PE values higher than 1.0 indicative of a dominance of *A. gambiae* s.s.

Similar dynamic patterns in sibling species composition of *A. gambiae* s.s. and *A. arabiensis* were described previously for the same area in western Kenya (White, 1972; Petrarca et al., 1991). However, these studies considered only adult and not larval populations. Minakawa et al. (1999) found that there was significant variation in the proportions of *A. gambiae* s.s and *A. arabiensis* between larval habitats in western Kenya, but this was on a spatial and not on a temporal scale. Gimnig et al. (2001) described the dynamics of both sibling species in the larval population in a

site approximately 80 km west of our study site, where environmental conditions are similar. However, this study focussed on describing the physical and chemical characteristics of the larval habitats of both species and concluded that there was no detectable segregation of *A. gambiae* s.s. and *A. arabiensis* into separate habitats. The results from our study presented in this paper show that both the larval and adult population are closely linked in numbers and species composition, with rainfall as the driving force of these dynamics.

The method used to sample larvae from aquatic habitats on the selected transect differed from previously described collection methods in this respect that only one single larva was collected from a larval habitat, whereas in other studies more than one larva was collected from a habitat (see, e.g. Gimnig et al. (2001)). We argue that bias in sibling species ratio was prevented in this way, since we assume that every larva collected represents the offspring of one individual adult female mosquito. However, it assumed that *A. gambiae* s.s. and *A. arabiensis* have equal chances of being collected from a site, but up to date there is no evidence that the two sibling species differ in this aspect of their behaviour. The availability of molecular markers (Zheng et al., 1996; Wang et al., 1999) makes it now possible to investigate to what extent larval habitats are shared between different families and to what extent one female distributes her eggs over different habitats. This, on its turn, makes it possible to investigate to what extent single larvae, collected with the described sampling method, represent the adult population.

The sampling method used did not allow for determination of sibling species composition within a larval habitat. Nonetheless, we know from other (unpublished) field data from the same area that the vast majority of habitats (>90%) was shared by *A. gambiae* s.s. and *A. arabiensis*. Our sampling method proved to be an accurate method to relate rainfall with the number of occupied larval habitats and adult mosquito densities, and to link temporal dynamics in sibling species ratio from the larval population with dynamics in this ratio from the adult population. The goals of the study and time and money considerations will determine the sampling method employed.

The temporal dynamics in sibling species composition will have its effect on the actual malaria risk in the area. A population consisting entirely of *A. gambiae* s.s. will involve a higher risk than a pure *A. arabiensis* population, because human biting rates (and sporozoite rate as a result) of the former species are higher than that of latter (Githeko et al., 1994). A sibling species composition pattern through time as observed in our study will involve a shift in the actual malaria risk. Data on adult *A. funestus* need to be included as well in order to estimate the additional malaria risk caused by this vector species.

Future studies of malaria epidemiology should consider whether the observed associations in the phenology and population dynamics of the aquatic and terrestrial stages of *A. gambiae* s.l. are characteristic throughout the distribution range of this important malaria vector. With this knowledge, a more complete picture will eventually be obtained of how future climatological changes affect the risk of malaria.

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